

# Perishable Lot-Sizing with Period-Dependent Shelf Lives

Compact Mixed-Integer Programming Model

## Indices and Sets

- $T = \{1, 2, \dots, \mathbf{T}\}$ : set of planning periods.
- $I$ : set of items.
- $i \in I$ : item index.
- $t \in T$ : production period.
- $u \in T$ : consumption period.

## Perishability

**Shelf life.** If item  $i$  is produced in period  $t$  and consumed in period  $u \geq t$ , its age at consumption is  $u - t$ . With a period-dependent shelf life  $m_{i,t} \geq 1$ , it is usable only if

$$u \leq t + m_{i,t} - 1.$$

**Consumption window from  $(i, t)$ .** For production in period  $t$ , the consumption window is

$$\{u \in T \mid t \leq u \leq \min(\mathbf{T}, t + m_{i,t} - 1)\}.$$

**Production coverage for  $(i, u)$ .** For demand in period  $u$ , feasible production periods are

$$\{t \in \{1, \dots, u\} \mid u \leq t + m_{i,t} - 1\}.$$

**Arc set.** Decision variables  $x_{i,t,u}$  are defined only when  $t \leq u \leq t + m_{i,t} - 1$ .

### Parameters

|                    |                                                                      |
|--------------------|----------------------------------------------------------------------|
| $d_{i,u}$          | demand of item $i$ in period $u$ .                                   |
| $c_i^{\text{var}}$ | variable production cost per unit.                                   |
| $h_i$              | unit holding cost per period (applied to age $u - t$ ).              |
| $s_{i,t}$          | setup cost in period $t$ for item $i$ (may be constant across $t$ ). |
| $b_i^{\text{var}}$ | backlog penalty per unit (only if backorders allowed).               |
| $\text{cap}_t$     | total production capacity in period $t$ .                            |
| $\text{cap}_{i,t}$ | optional per-item capacity in period $t$ .                           |
| $m_{i,t}$          | shelf life for production of $i$ in $t$ .                            |
| $W$                | optional warehouse capacity (end-of-period inventory limit).         |
| $M_i$              | tight big- $M$ for setups, $M_i = \sum_{u \in T} d_{i,u}$ .          |

### Decision Variables

|                                     |                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------|
| $x_{i,t,u} \in \mathbb{Z}_{\geq 0}$ | quantity of $i$ produced in $t$ and consumed in $u$ , only if $t \leq u \leq t + m_{i,t} - 1$ . |
| $y_{i,t} \in \{0, 1\}$              | setup indicator for $i$ in period $t$ .                                                         |
| $b_{i,u} \in \mathbb{Z}_{\geq 0}$   | backlog at end of period $u$ for $i$ (only if backorders are used).                             |

### Per-Unit Cost

The total per-unit cost for moving one unit of  $i$  from  $t$  to  $u$  is

$$C_{i,t,u} = c_i^{\text{var}} + h_i(u - t).$$

This covers production and holding. Setup and backlog are modelled separately:

$$\text{Setup cost at } (i, t) : s_{i,t}, \quad \text{Backlog penalty at } (i, u) : b_i^{\text{var}}.$$

### Objective

$$\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{t \in T} \sum_{\substack{u \in T \\ t \leq u \leq t + m_{i,t} - 1}} C_{i,t,u} x_{i,t,u} \\
& + \sum_{i \in I} \sum_{t \in T} s_{i,t} y_{i,t} \\
& + (\text{only if backorders allowed}) \sum_{i \in I} \sum_{u \in T} b_i^{\text{var}} b_{i,u}.
\end{aligned} \tag{1}$$

### Constraints

**Global capacity:** For each  $t \in \{1, \dots, T\}$ ,

$$\sum_{i \in I} \sum_{\substack{u \in T \\ t \leq u \leq t + m_{i,t} - 1}} x_{i,t,u} \leq \text{cap}_t.$$

**Per-item capacity (if given):** For each  $i \in I$ ,  $t \in T$ ,

$$\sum_{\substack{u \in T \\ t \leq u \leq t+m_{i,t}-1}} x_{i,t,u} \leq \text{cap}_{i,t}.$$

**Setup linking:** For each  $i \in I$ ,  $t \in T$ ,

$$\sum_{\substack{u \in T \\ t \leq u \leq t+m_{i,t}-1}} x_{i,t,u} \leq M_i y_{i,t}.$$

**Warehouse capacity (if given):** For each  $u \in \{1, \dots, T-1\}$ ,

$$\sum_{i \in I} \sum_{t=1}^u \sum_{\substack{w \in T \\ t \leq w \leq t+m_{i,t}-1 \\ w > u}} x_{i,t,w} \leq W.$$

**Demand balance (no backorders):** For each  $i \in I$ ,  $u \in T$ ,

$$\sum_{\substack{t=1, \dots, u \\ u \leq t+m_{i,t}-1}} x_{i,t,u} = d_{i,u}.$$

**Demand balance (with backorders):** For  $u = 1$ :

$$\sum_{\substack{t=1 \\ 1 \leq t+m_{i,t}-1}} x_{i,t,1} + b_{i,1} = d_{i,1}.$$

For  $u \geq 2$ :

$$\sum_{\substack{t=1, \dots, u \\ u \leq t+m_{i,t}-1}} x_{i,t,u} + b_{i,u} = d_{i,u} + b_{i,u-1}.$$